

6221 tutor

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Exam overview

5 topics

Kinematics

Dynamics and Control

Mobile Robots

Structure and Pose Estimation

Kalman Filter and Sensor Integration

1. kinematics

Kinematics

Problem definition:

a) link diagram using D-H algorithm

b) Kinematics parameters

or

b) transformation matrix

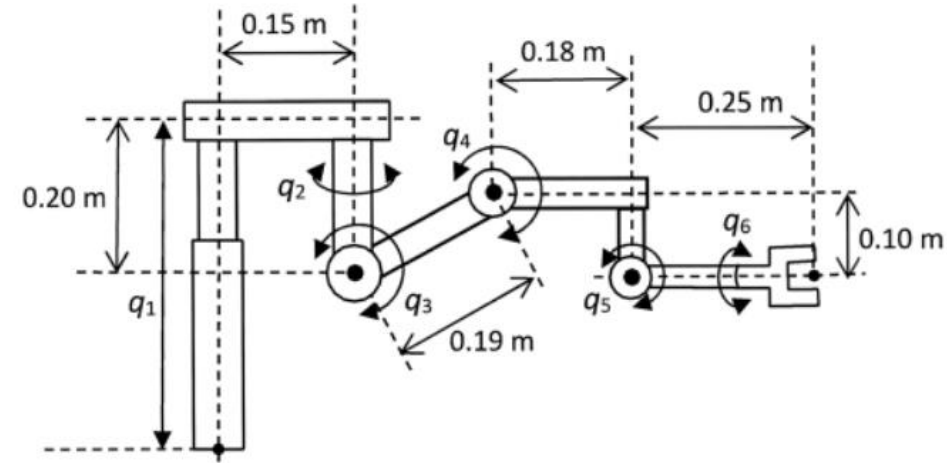


Figure 1

(a) Obtain the link coordinate diagram by using the Denavit-Hartenberg (D-H) algorithm.

(12 Marks)

(b) Derive the kinematic parameters of the robot based on the coordinate diagram obtained in part (a).

(8 Marks)

$$T_{K-1}^K = \begin{bmatrix} \cos \theta_K & -\cos \alpha_K \sin \theta_K & \sin \alpha_K \sin \theta_K & a_K \cos \theta_K \\ \sin \theta_K & \cos \alpha_K \cos \theta_K & -\sin \alpha_K \cos \theta_K & a_K \sin \theta_K \\ 0 & \sin \alpha_K & \cos \alpha_K & d_K \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Kinematics – principle

The four parameters describe the transformation from the coordinate system $k-1$ attached to link $k-1$ to the coordinate system k attached to link k . The **origin** of coordinate system $k-1$ is O_{k-1} , and the **coordinate** axes are $(x_{k-1}, y_{k-1}, z_{k-1})$; the origin of coordinate system k is O_k , and the coordinate axes are (x_k, y_k, z_k)

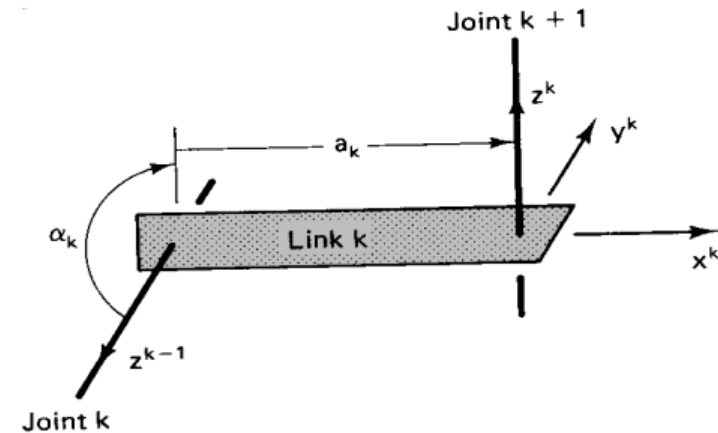
1. Z^K : each Z always align with the axis of joint
2. origin^K : intersection of Z s, or norm of Z s if parallel
3. X^K : to be orthogonal to Z s. away from prev Z
4. Y^K : right handed

as for the tool tip:

Z : r3 approach

Y : r2 sliding

X : r1 normal



Kinematics – principle

- θ_k is defined as the angle between x^{k-1} and x^k axes about the z^{k-1} axis.
- d_k is the distance between x^{k-1} and x^k axes **along** z^{k-1} axis.
- a_k is the distance between z^{k-1} and z^k axes **along** x^k axis.
- α_k is the angle between z^{k-1} and z^k axes **about** x^k axis.

	Name	Meaning	revolute joints	prismatic joints
θ_k	Joint Angle	The amount of rotation about the joint axis	Variable	Fix
d_k	Joint distance	The amount of offset along the previous joint axis	Fixed	Variable
a_k	Link Length	The shortest distance (common perpendicular length) between joint axes z_{k-1} and z_k	Fixed	Fixed
α_k	Link Twist Angle	The degree of rotation between adjacent joint axes	Fixed	Fixed

Kinematics – principle

Illustrating Example:

Algorithm: D-H Representation

Number the joints from 1 to n starting with the base and ending with the tool yaw, pitch, and roll, in that order.

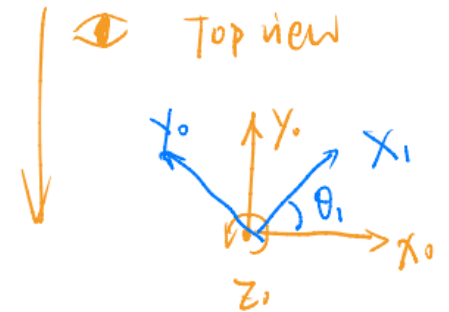
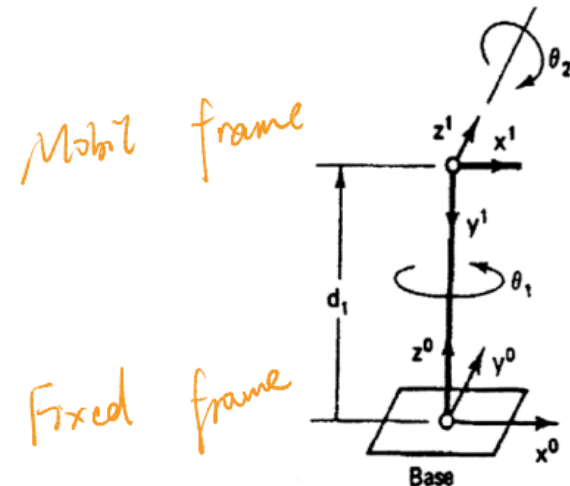
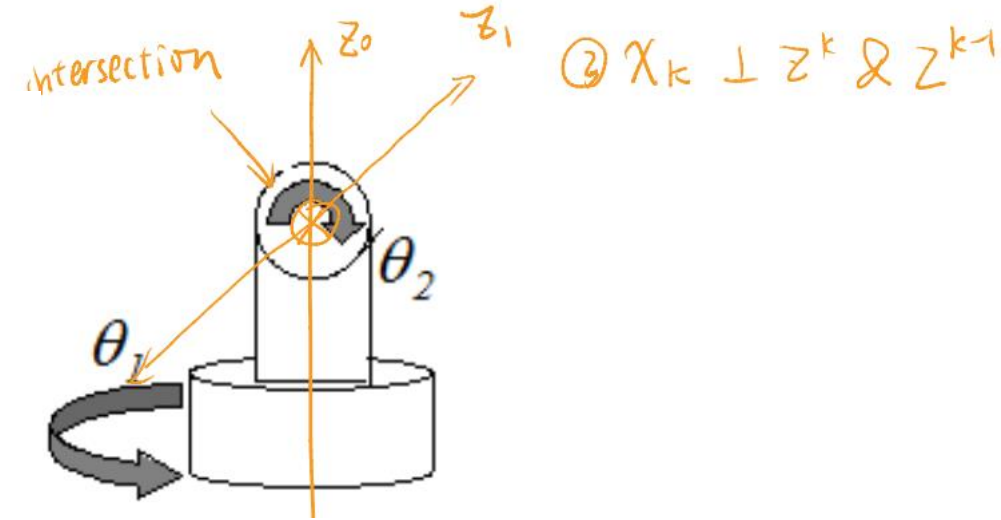
Assign a right-handed orthonormal coordinate frame L_0 to the robot base, making sure that z^0 aligns with the axis of joint 1. Set $k=1$.

x, y free to assign.
follow right hand rule

1. The z^k axis aligns with the axis of joint $k+1$.
2. The intersection of z^k and z^{k-1} axes is selected as the origin of L_k . If they do not intersect, use the intersection of z^k with a common normal between z^k and z^{k-1} .
3. Select x^k to be orthogonal to both z^k and z^{k-1} . If z^k and z^{k-1} are parallel, point x^k away from z^{k-1} . $\textcircled{3} x_k \perp z^k \& z^{k-1}$
4. Select y^k to form a right-handed orthonormal coordinate frame L_k .

Set $k = k+1$. If $k < n$, go to step 1; else, continue.

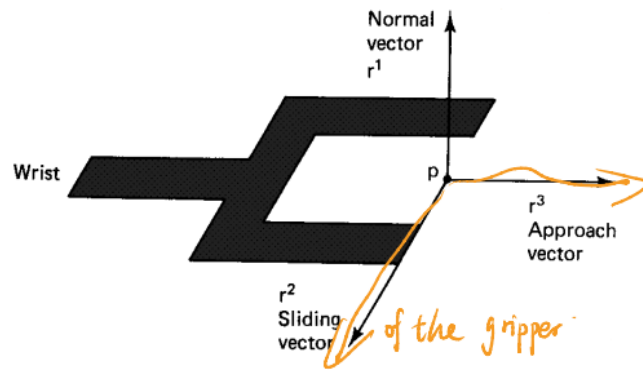
Set the origin of L_n at the tool tip. Align z^n with the approach vector, y^n with the sliding vector, and x^n with the normal vector of the tool.



Kinematics – principle

Tool tips

2.5.2 Normal, Sliding, and Approach Vectors



The orientation of the tool can be expressed in rectangular coordinates by a rotation matrix $R = [r^1, r^2, r^3]$, where the three columns of R correspond to the

- normal vector,
- sliding vector, and
- approach vectors,

respectively. The origin of the $\{r^1, r^2, r^3\}$ frame is usually placed at the tool tip.

$$T_{k-1}^k = \begin{pmatrix} C\theta_k & -C\alpha_k S\theta_k & S\alpha_k S\theta_k & a_k C\theta_k \\ S\theta_k & C\alpha_k C\theta_k & -S\alpha_k C\theta_k & a_k S\theta_k \\ 0 & S\alpha_k & C\alpha_k & d_k \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

provide by exam

$$T_{k-1}^k(\theta_k, d_k, a_k, \alpha_k) = \overset{I}{Rot}(\theta_k, 3) Tran(d_k i^3) Tran(a_k i^1) Rot(\alpha_k, 1)$$

$$\begin{pmatrix} \cos \theta_k & -\sin \theta_k & 0 & 0 \\ \sin \theta_k & \cos \theta_k & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} : \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_k \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & a_k \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha_k & -\sin \alpha_k & 0 \\ 0 & \sin \alpha_k & \cos \alpha_k & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The **transformation** from coordinate system k to coordinate system $k-1$, denoted as T_{k-1}^k , is obtained by combining four basic transformations associated with the D-H parameters $(\theta_k, d_k, a_k, \alpha_k)$.

Kinematics – practise1

1. A robotic manipulator with six joints is shown in Figure 1.

23-24 S2 Q1

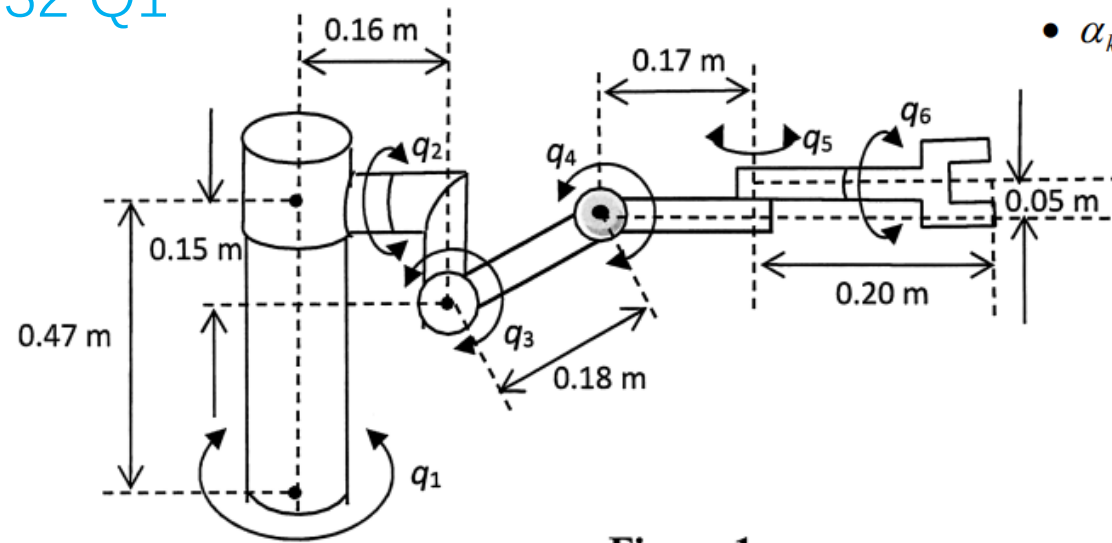


Figure 1

- θ_k is defined as the angle between x^{k-1} and x^k axes about the z^{k-1} axis.
- d_k is the distance between x^{k-1} and x^k axes **along** z^{k-1} axis.
- a_k is the distance between z^{k-1} and z^k axes **along** x^k axis.
- α_k is the angle between z^{k-1} and z^k axes **about** x^k axis.

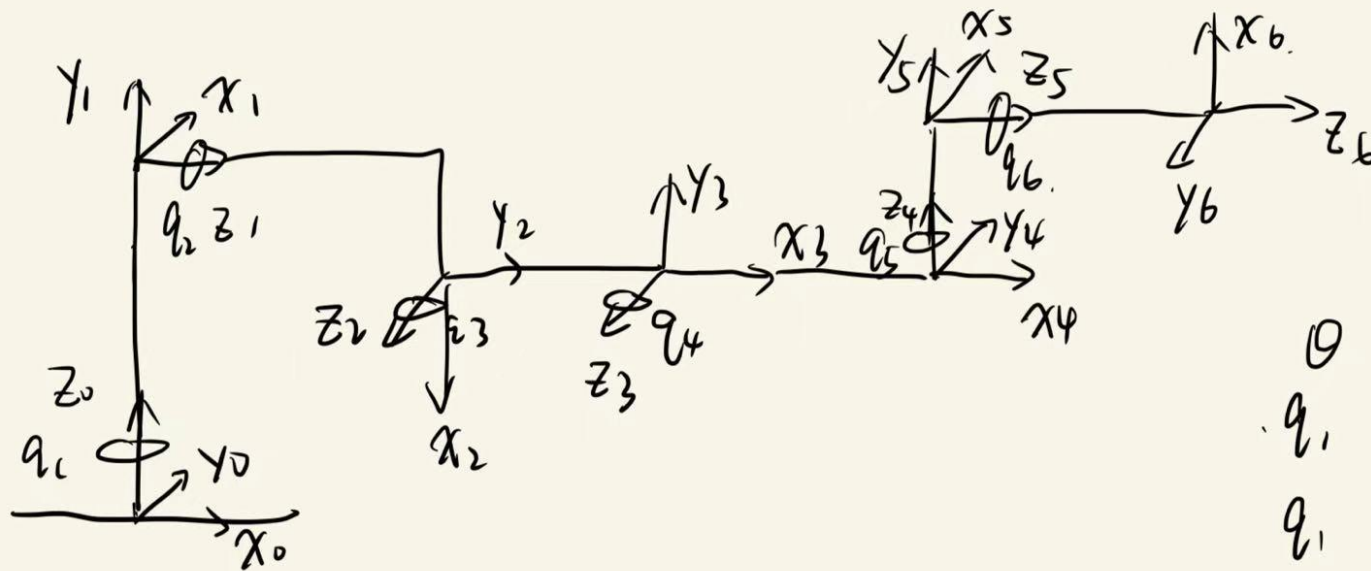
(a) Obtain the link coordinate diagram by using the Denavit-Hartenberg (D-H) algorithm.

(12 Marks)

(b) Derive the kinematic parameters of the robot based on the coordinate diagram obtained in part (a).

(8 Marks)

Kinematics – practise1



	θ	d	a	α
q_1		0.47	0	$\frac{\pi}{2}$
q_2		0.16	0.15	$\frac{\pi}{2}$
q_3		0	0.18	0
q_4		0	0.17	$-\frac{\pi}{2}$
q_5		0.05	0	$\frac{\pi}{2}$
q_6		0.2	0	0

Kinematics – practise2

- θ_k is defined as the angle between x^{k-1} and x^k axes about the z^{k-1} axis.
- d_k is the distance between x^{k-1} and x^k axes **along** z^{k-1} axis.
- a_k is the distance between z^{k-1} and z^k axes **along** x^k axis.
- α_k is the angle between z^{k-1} and z^k axes **about** x^k axis.

1. A robotic manipulator with seven joints is shown in Figure 1.

22-23 S2 Q1

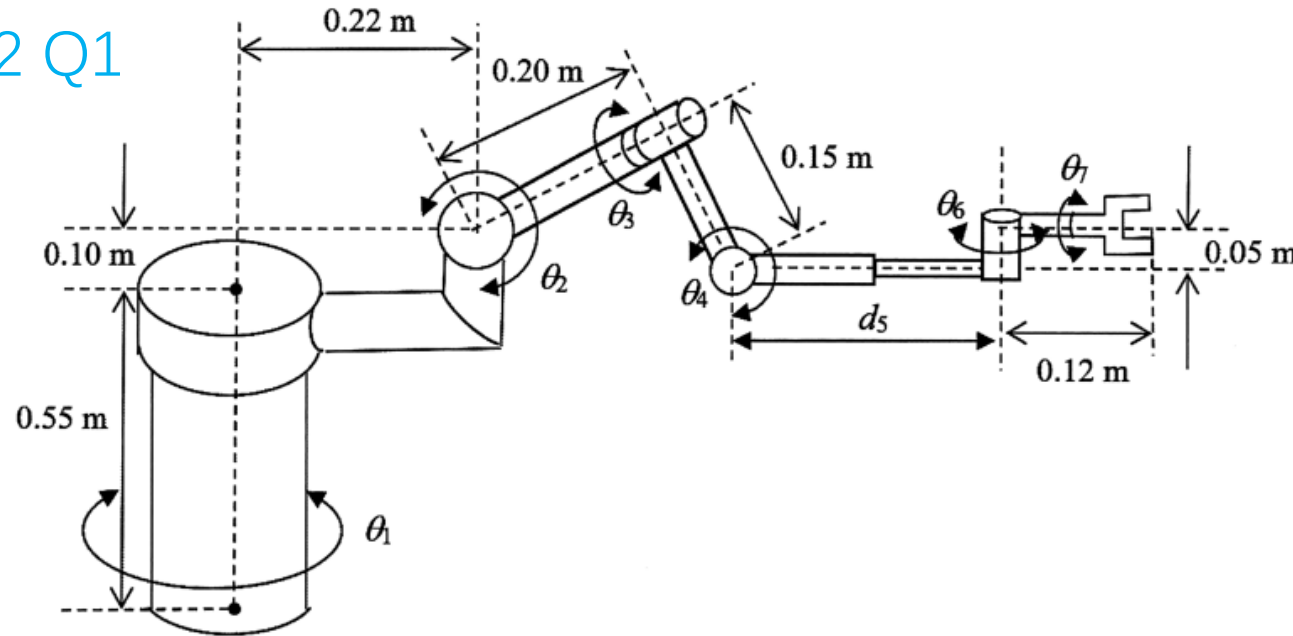


Figure 1

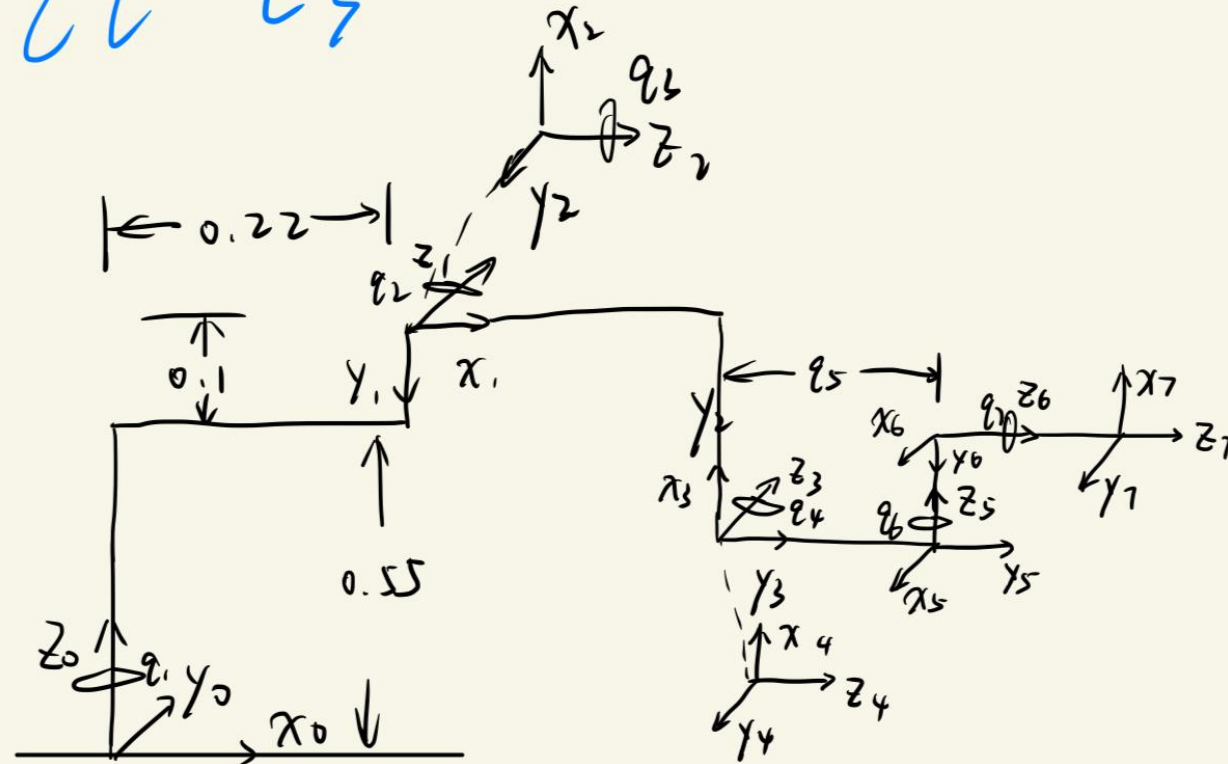
(a) Obtain the link coordinate diagram by using the Denavit-Hartenberg (D-H) algorithm.

(12 Marks)

(b) Derive the kinematic parameters of the robot based on the coordinate diagram obtained in part (a).

Kinematics – practise2

22-23



	θ	d	a	α
q_1	0.65	0.22	$\frac{\pi}{2}$	
q_2	0	0	$-\frac{\pi}{2}$	
q_3	0.2	0.15	$\frac{\pi}{2}$	
q_4	0	0	$-\frac{\pi}{2}$	
$\frac{\pi}{2}$	q_5	0	$\frac{\pi}{2}$	
q_6	0.05	0	$-\frac{\pi}{2}$	
q_7	0.12	0	0	

Kinematics – practise3

- θ_k is defined as the angle between x^{k-1} and x^k axes about the z^{k-1} axis.
- d_k is the distance between x^{k-1} and x^k axes **along** z^{k-1} axis.
- a_k is the distance between z^{k-1} and z^k axes **along** x^k axis.
- α_k is the angle between z^{k-1} and z^k axes **about** x^k axis.

1. A robotic manipulator with six joints is mounted on a wall as shown in Figure 1.

24-25 S2 Q1

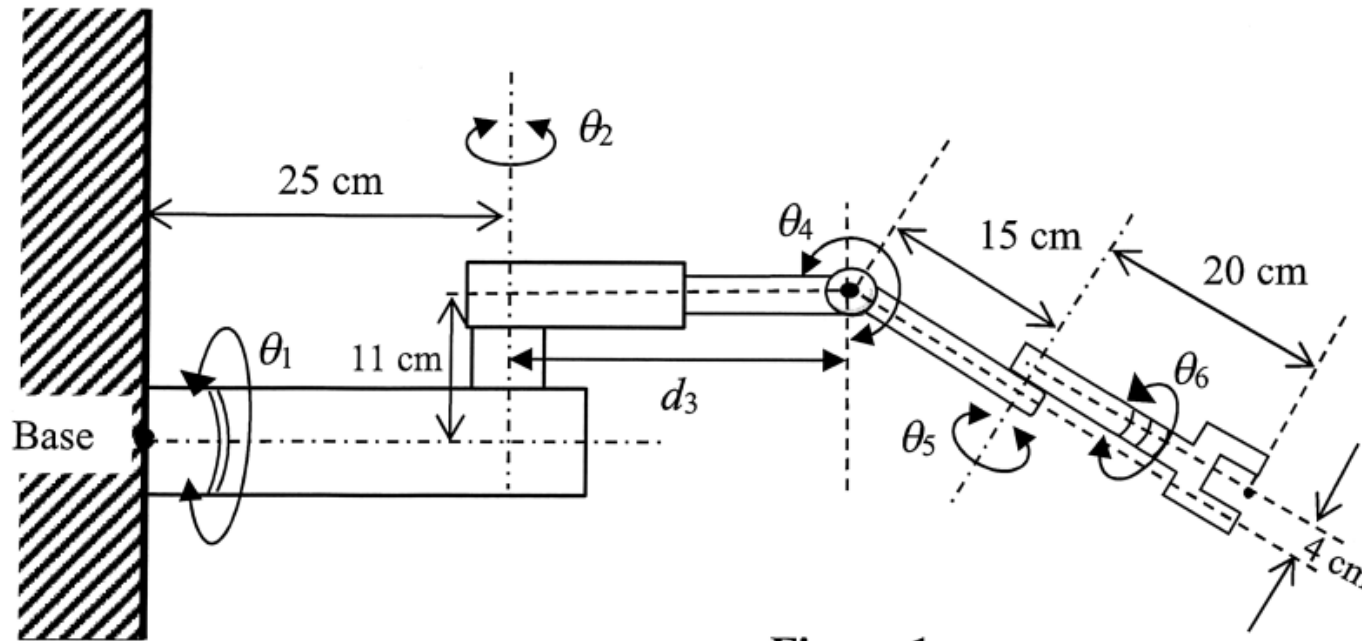
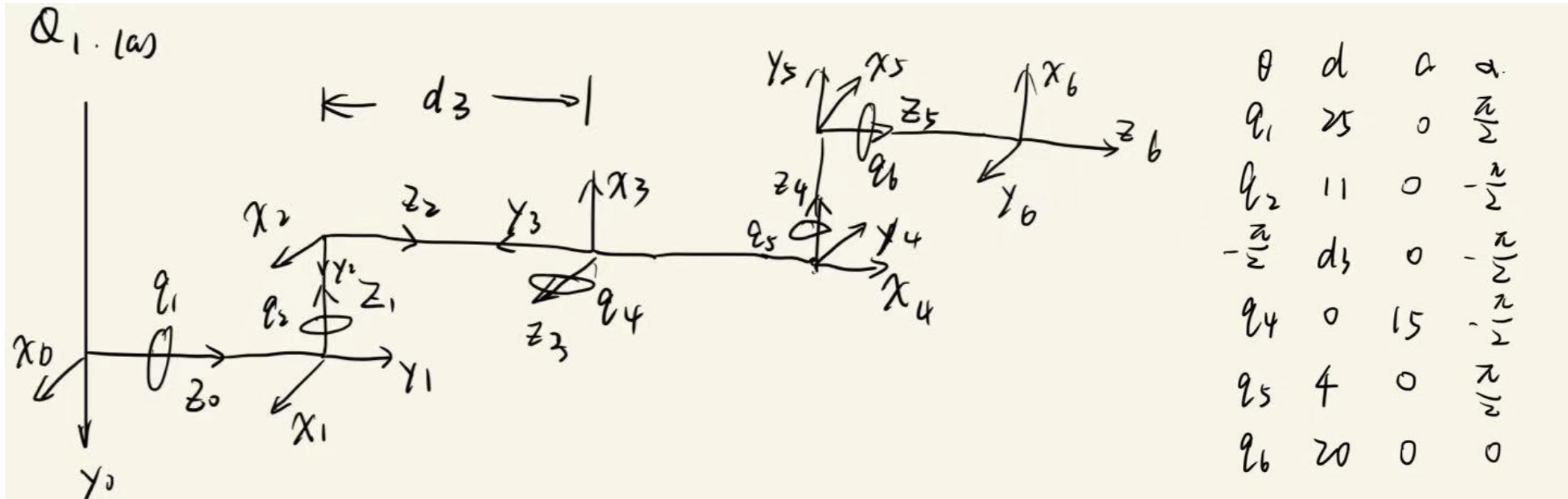


Figure 1

- (a) Obtain the link coordinate diagram by using the Denavit-Hartenberg (D-H) algorithm.

(11 Marks)

Kinematics – practise3



Kinematics – practise3

24-25 S2 Q1

- (b) Use the link coordinate diagram obtained in part (a) to derive and tabulate the kinematic parameters of the robot. Compute the link-coordinate homogeneous transformation matrices of the first 3 joints. The link-coordinate homogeneous transformation matrix is given as:

$$T_{K-1}^K = \begin{bmatrix} \cos \theta_K & -\cos \alpha_K \sin \theta_K & \sin \alpha_K \sin \theta_K & a_K \cos \theta_K \\ \sin \theta_K & \cos \alpha_K \cos \theta_K & -\sin \alpha_K \cos \theta_K & a_K \sin \theta_K \\ 0 & \sin \alpha_K & \cos \alpha_K & d_K \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where θ_K , d_K , a_K , α_K are the kinematic parameters. (Note: multiplication of the matrices is not required).

(9 Marks)

θ	d	a	α
q_1	25	0	$\frac{\pi}{2}$
q_2	11	0	$-\frac{\pi}{2}$
$-\frac{\pi}{2}$	d_3	0	$-\frac{\pi}{2}$
q_4	0	15	$-\frac{\pi}{2}$
q_5	4	0	$\frac{\pi}{2}$
q_6	20	0	0

Kinematics – practise3

$$T_{K-1}^K = \begin{bmatrix} \cos \theta_K & -\cos \alpha_K \sin \theta_K & \sin \alpha_K \sin \theta_K & a_K \cos \theta_K \\ \sin \theta_K & \cos \alpha_K \cos \theta_K & -\sin \alpha_K \cos \theta_K & a_K \sin \theta_K \\ 0 & \sin \alpha_K & \cos \alpha_K & d_K \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

cb) $T_0^1 = \begin{bmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & 25 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$T_1^2 = \begin{bmatrix} c_2 & 0 & -s_2 & 0 \\ s_2 & 0 & c_2 & 0 \\ 0 & -1 & 0 & 11 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2^3 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

θ	d	a	α
q_1	25	0	$\frac{\pi}{2}$
q_2	11	0	$-\frac{\pi}{2}$
$-\frac{\pi}{2}$	d_3	0	$-\frac{\pi}{2}$
q_4	0	15	$-\frac{\pi}{2}$
q_5	4	0	$\frac{\pi}{2}$
q_6	20	0	0

Inverse Kinematics

Problem definition:

- a) tool configuration Jacobian matrix.
- b) Solve kinematic problem
- sometimes
- c) Determine the approach vector

A robot manipulator with four joint variables is mounted on a mobile platform. The transformation matrix from tool tip to base coordinate of the robot is given as:

$$T_{base}^{tool} = \begin{bmatrix} -S_1 S_3 C_4 - C_1 S_4 & -S_1 S_3 S_4 - C_1 C_4 & -S_1 C_3 & S_1 q_2 + 0.25 S_1 C_3 \\ C_1 S_3 C_4 - S_1 S_4 & -C_1 C_3 S_4 - S_1 C_4 & C_1 C_3 & C_1 q_2 + 0.25 C_1 C_3 \\ -C_3 C_4 & -C_3 S_4 & -S_3 & -0.25 S_3 + 0.25 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where q_1, q_2, q_3 are the joint variables for the major axes, q_4 is the tool roll angle, $C_k = \cos q_k$ and $S_k = \sin q_k$.

- (i) Derive the tool configuration Jacobian matrix of the manipulator.
- (ii) Given that $x = 0.2, y = 0.2, z = 0$, solve the inverse kinematic problem to obtain q_1, q_2, q_3 . (Note: orientation is not required).
- (iii) Determine the approach vector of the robot at this joint configuration.

Definition: Tool-configuration Vector.

Let p and R denote the position and orientation of the tool frame relative to the base frame where q_n represents the tool roll angle. Then the *tool-configuration vector* is a vector w in R^6 defined:

$$w = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \\ w_6 \end{pmatrix} = \begin{pmatrix} p \\ \exp(q_n / \pi) r^3 \end{pmatrix} \in R^6$$

- The first three components p , represent the tool-tip position,
- The last three components, $\exp(q_n / \pi) r^3$, represent the tool orientation.

Inverse Kinematics

tool configuration matrix

$$w = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \\ w_6 \end{pmatrix}$$

$$= \begin{pmatrix} p \\ \exp(q_n / \pi) r^3 \end{pmatrix}$$

$$T_{base}^{tool} = \begin{bmatrix} -S_1 S_3 C_4 - C_1 S_4 & -S_1 S_3 S_4 - C_1 C_4 & -S_1 C_3 & S_1 q_2 + 0.25 S_1 C_3 \\ C_1 S_3 C_4 - S_1 S_4 & -C_1 C_3 S_4 - S_1 C_4 & C_1 C_3 & C_1 q_2 + 0.25 C_1 C_3 \\ -C_3 C_4 & -C_3 S_4 & -S_3 & -0.25 S_3 + 0.25 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Find the Arm Equation (page 72).

$$T_{base}^{tool}(q) = \begin{pmatrix} R(q) & p(q) \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- 3x3 matrix $R(q)$ specifies the **orientation** of the tool, while the 3x1 matrix
- $p(q)$ specifies the **position** of the tool tip.

$$w = \begin{bmatrix} S_1 q_2 + 0.25 S_1 C_3 \\ C_1 q_2 + 0.25 C_1 C_3 \\ -0.25 S_3 + 0.25 \\ \exp(q_4 / \pi) (-S_1 C_3) \\ \exp(q_4 / \pi) (C_1 C_3) \\ \exp(q_4 / \pi) (-S_3) \end{bmatrix}$$

Inverse Kinematics

The first three components of $w(q)$ represent the tool-tip position, while the last three components of $w(q)$ represent the approach vector of the tool scaled by $\exp(q_5 / \pi)$, where q_5 is the tool roll angle.

If we differentiate $w(q)$, this yields the following tool Jacobian matrix, where $v^k(q)$ denotes the k th column of $V(q)$ for $1 \leq k \leq 5$:

The column i of tool configuration Jacobian matrix of the manipulator is

$$v^i(q) = \frac{\partial w}{\partial q_i}$$

Inverse Kinematics

$$w = \begin{bmatrix} S_1 q_2 + 0.25 S_1 C_3 \\ C_1 q_2 + 0.25 C_1 C_3 \\ -0.25 S_3 + 0.25 \\ \exp(q_4/\pi)(-S_1 C_3) \\ \exp(q_4/\pi)(C_1 C_3) \\ \exp(q_4/\pi)(-S_3) \end{bmatrix}$$

$$v^i(q) = \frac{\partial w}{\partial q_i}$$

$$v^1(q) = \begin{bmatrix} C_1 q_2 + 0.25 C_1 C_3 \\ -S_1 q_2 - 0.25 S_1 C_3 \\ 0 \\ \exp(q_4/\pi)(-C_1 C_3) \\ \exp(q_4/\pi)(-S_1 C_3) \\ 0 \end{bmatrix} \quad v^2(q) = \begin{bmatrix} S_1 \\ C_1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$v^3(q) = \begin{bmatrix} -0.25 S_1 S_3 \\ -0.25 C_1 S_3 \\ -0.25 C_3 \\ \exp(q_4/\pi)(S_1 S_3) \\ \exp(q_4/\pi)(-C_1 S_3) \\ \exp(q_4/\pi)(-C_3) \end{bmatrix} \quad v^4(q) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \exp(q_4/\pi)(-S_1 C_3)/\pi \\ \exp(q_4/\pi)(C_1 C_3)/\pi \\ \exp(q_4/\pi)(-S_3)/\pi \end{bmatrix}$$

A robot manipulator with four joint variables is mounted on a mobile platform. The transformation matrix from tool tip to base coordinate of the robot is given as:

$$T_{base}^{tool} = \begin{bmatrix} -S_1 S_3 C_4 - C_1 S_4 & -S_1 S_3 S_4 - C_1 C_4 & -S_1 C_3 & S_1 q_2 + 0.25 S_1 C_3 \\ C_1 S_3 C_4 - S_1 S_4 & -C_1 C_3 S_4 - S_1 C_4 & C_1 C_3 & C_1 q_2 + 0.25 C_1 C_3 \\ -C_3 C_4 & -C_3 S_4 & -S_3 & -0.25 S_3 + 0.25 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where q_1, q_2, q_3 are the joint variables for the major axes, q_4 is the tool roll angle, $C_k = \cos q_k$ and $S_k = \sin q_k$.

- Derive the tool configuration Jacobian matrix of the manipulator.
- Given that $x = 0.2, y = 0.2, z = 0$, solve the inverse kinematic problem to obtain q_1, q_2, q_3 . (Note: orientation is not required).
- Determine the approach vector of the robot at this joint configuration.

$$(b) \quad W(q) = \begin{bmatrix} S_1 q_2 + 0.25 S_1 C_3 \\ C_1 q_2 + 0.25 C_1 C_3 \\ -0.25 S_3 + 0.25 \\ -\exp(q_4/\pi) \cdot S_1 C_3 \\ \exp(q_4/\pi) \cdot C_1 C_3 \\ -\exp(q_4/\pi) \cdot S_3 \end{bmatrix}$$

$$J(q) = \begin{bmatrix} \frac{\partial W}{\partial q_1} & \frac{\partial W}{\partial q_2} & \frac{\partial W}{\partial q_3} & \frac{\partial W}{\partial q_4} \end{bmatrix}$$

$$= \begin{bmatrix} C_1 q_2 + 0.25 C_1 C_3 & S_1 & -0.25 S_1 S_3 & 0 \\ -S_1 q_2 - 0.25 S_1 C_3 & C_1 & -0.25 C_1 S_3 & 0 \\ 0 & 0 & -0.25 C_3 & 0 \\ -\exp(q_4/\pi) C_1 C_3 & 0 & \exp(q_4/\pi) S_1 S_3 & -\exp(q_4/\pi) S_1 C_3 / \pi \\ -\exp(q_4/\pi) S_1 C_3 & 0 & -\exp(q_4/\pi) C_1 S_3 & \exp(q_4/\pi) C_1 C_3 / \pi \\ 0 & 0 & -\exp(q_4/\pi) C_3 & -\exp(q_4/\pi) S_3 / \pi \end{bmatrix}$$

Inverse Kinematics

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where q_1, q_2, q_3 are the joint variables for the major axes, q_4 is the tool roll angle, $C_k = \cos q_k$ and $S_k = \sin q_k$.

- (i) Derive the tool configuration Jacobian matrix of the manipulator.
- (ii) Given that $x = 0.2, y = 0.2, z = 0$, solve the inverse kinematic problem to obtain q_1, q_2, q_3 . (Note: orientation is not required).
- (iii) Determine the approach vector of the robot at this joint configuration.

Definition: Tool-configuration Vector.

Let p and R denote the position and orientation of the tool frame relative to the base frame where q_n represents the tool roll angle. Then the *tool-configuration vector* is a vector w in R^6 defined:

Inverse Kinematics

$$w = \begin{bmatrix} S_1 q_2 + 0.25 S_1 C_3 \\ C_1 q_2 + 0.25 C_1 C_3 \\ -0.25 S_3 + 0.25 \\ \exp(q_4/\pi)(-S_1 C_3) \\ \exp(q_4/\pi)(C_1 C_3) \\ \exp(q_4/\pi)(-S_3) \end{bmatrix}$$

(ii) Obtain q_1, q_2, q_3 .

$$\begin{cases} x = 0.2 = S_1 q_2 + 0.25 S_1 C_3 \\ y = 0.2 = C_1 q_2 + 0.25 C_1 C_3 \\ z = 0 = -0.25 S_3 + 0.25 \end{cases} \implies \begin{cases} q_1 = \pi/4 \\ q_2 = \sqrt{2}/5 \\ q_3 = \pi/2 \end{cases}$$

A robot manipulator with four joint variables is mounted on a mobile platform. The transformation matrix from tool tip to base coordinate of the robot is given as:

$$T_{base}^{tool} = \begin{bmatrix} -S_1 S_3 C_4 - C_1 S_4 & -S_1 S_3 S_4 - C_1 C_4 & -S_1 C_3 & S_1 q_2 + 0.25 S_1 C_3 \\ C_1 S_3 C_4 - S_1 S_4 & -C_1 S_3 S_4 - S_1 C_4 & C_1 C_3 & C_1 q_2 + 0.25 C_1 C_3 \\ -C_3 C_4 & -C_3 S_4 & -S_3 & -0.25 S_3 + 0.25 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where q_1, q_2, q_3 are the joint variables for the major axes, q_4 is the tool roll angle, $C_k = \cos q_k$ and $S_k = \sin q_k$.

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Inverse Kinematics

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A robot manipulator with four joint variables is mounted on a mobile platform. The transformation matrix from tool tip to base coordinate of the robot is given as:

$$T_{base}^{tool} = \begin{bmatrix} -S_1 S_3 C_4 - C_1 S_4 & -S_1 S_3 S_4 - C_1 C_4 & -S_1 C_3 & S_1 q_2 + 0.25 S_1 C_3 \\ C_1 S_3 C_4 - S_1 S_4 & -C_1 S_3 S_4 - S_1 C_4 & C_1 C_3 & C_1 q_2 + 0.25 C_1 C_3 \\ -C_3 C_4 & -C_3 S_4 & -S_3 & -0.25 S_3 + 0.25 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where q_1, q_2, q_3 are the joint variables for the major axes, q_4 is the tool roll angle, $C_k = \cos q_k$ and $S_k = \sin q_k$.

- (i) Derive the tool configuration Jacobian matrix of the manipulator.
- (ii) Given that $x = 0.2, y = 0.2, z = 0$, solve the inverse kinematic problem to obtain q_1, q_2, q_3 . (Note: orientation is not required).
- (iii) Determine the approach vector of the robot at this joint configuration.

(iii) Determine the approach vector of the robot at this joint configuration.

$$r^3 = \begin{bmatrix} -S_1 C_3 \\ C_1 C_3 \\ -S_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

22-23 S2 Q3

- (b) A robot manipulator with three joint variables q_1, q_2, q_3 are mounted on a mobile robot. The link-coordinate homogeneous transformation matrix from the base coordinate to the tool coordinate of the robotic manipulator is given as:

$$T_{base}^{tool} = \begin{bmatrix} C_1 C_2 & -S_1 & C_1 S_2 & C_1 S_2 (q_3 + 0.1) \\ S_1 C_2 & C_1 & S_1 S_2 & S_1 S_2 (q_3 + 0.1) \\ -S_2 & 0 & C_2 & C_2 (q_3 + 0.1) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where $S_1 = \sin(q_1)$, $S_2 = \sin(q_2)$, $C_1 = \cos(q_1)$, $C_2 = \cos(q_2)$.

- (i) Solve the inverse kinematic problem using an analytic method to express $(q_1, q_2, q_3)^T$ in terms of the position of the end effector $(x, y, z)^T$. (Note: orientation is not required).
- (ii) If the robot is used to pick up an object with a centroid given as $x = 0.15$, $y = 0.25$, $z = 0.025$ units, calculate the joint configuration. Discuss the problems associated with this task.

$$T_{base}^{tool} = \begin{bmatrix} C_1 C_2 & -S_1 & C_1 S_2 & C_1 S_2 (q_3 + 0.1) \\ S_1 C_2 & C_1 & S_1 S_2 & S_1 S_2 (q_3 + 0.1) \\ -S_2 & 0 & C_2 & C_2 (q_3 + 0.1) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where $S_1 = \sin(q_1)$, $S_2 = \sin(q_2)$, $C_1 = \cos(q_1)$, $C_2 = \cos(q_2)$.

- (i) Solve the inverse kinematic problem using an analytic method to express $(q_1, q_2, q_3)^T$ in terms of the position of the end effector $(x, y, z)^T$. (Note: orientation is not required).
- (ii) If the robot is used to pick up an object with a centroid given as $x = 0.15$, $y = 0.25$, $z = 0.025$ units, calculate the joint configuration. Discuss the problems associated with this task.

$$\begin{cases} C_1 S_2 (q_3 + 0.1) = x \\ S_1 S_2 (q_3 + 0.1) = y \\ C_2 (q_3 + 0.1) = z \end{cases}$$

2. Dynamics and Control

Dynamics and Control

The dynamic equations of a robot, which is in contact with a frictionless surface, are given as follows:

$$\begin{aligned}(3 + 2 \cos(q_2))\ddot{q}_1 + (1 + \cos(q_2))\ddot{q}_2 + (\dot{q}_1^2 + 2\dot{q}_1\dot{q}_2) \sin(q_2) + 3\dot{q}_1 + 2g \cos(q_1) \\ + g \cos(q_1 + q_2) = u_1 \\ (1 + \cos(q_2))\ddot{q}_1 + \ddot{q}_2 + 2 \sin(q_2)\dot{q}_1^2 + 5\dot{q}_2 + g \cos(q_1 + q_2) = u_2 \\ 2\ddot{q}_3 + 7\dot{q}_3 + 2g + f = u_3\end{aligned}$$

where q_1, q_2, q_3 are the joint variables, u_1, u_2, u_3 are the control inputs and $g = 9.8 \text{ m/s}^2$ is the acceleration due to gravity. The first two joints possess unmodelled resonance at 12 rad/sec and the third joint possesses unmodelled resonance at 16 rad/sec. The contact force exerted on the environment is given by:

$$f = 10(q_3 - 0.1).$$

- (a) Design a hybrid position and force controller for the robot so that the first two joints are critically damped, and the third joint is overdamped with a damping ratio of 1.2. The gains should be as large as possible, and the system should not excite all the unmodelled resonances.

(14 Marks)

- (b) The controller designed in part (a) is now implemented on the robot in a space station, without any modification. Explain the possible effects and derive the error equations.

(6 Marks)

The dynamic equations of a robot when it is in contact with a workpiece are given as follows:

$$\begin{aligned}(5q_2^2 + 1)\ddot{q}_1 + 10q_2\dot{q}_1\dot{q}_2 + 3\dot{q}_1 + 47.5 q_2 \cos(q_1) + d_1 = u_1 \\ 5\ddot{q}_2 - 5q_2\dot{q}_1^2 + 5\dot{q}_2 + 47.5 \sin(q_1) + d_2 = u_2 \\ 3\ddot{q}_3 + 2\dot{q}_3 + 29.4 + d_3 + f = u_3\end{aligned}$$

where u_1, u_2, u_3 are the control inputs, q_1, q_2, q_3 are the joint variables, d_1, d_2, d_3 are the unknown disturbances and $f = 22(q_3 - 0.2)$ is the contact force. The system possesses unmodelled resonances at 5 rad/s, 10 rad/s and 15 rad/s.

- (a) If d_1, d_2, d_3 are zero, design a hybrid position and force controller for the robot to track the desired trajectories q_{1d}, q_{2d} and the desired force f_d . The motion control subspace should be overdamped with a damping ratio of 1.05, and the force control subspace should be critically damped. The control system should not excite all the unmodelled resonances.

(14 Marks)

- (b) If d_1, d_2, d_3 are not zero, derive the error equations of the system based on the hybrid position and force controller designed in part (a).

(6 Marks)

Dynamic Model

- For a robot with n degrees of freedom (DOF), its dynamic equations can usually be written as:

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + g(\theta) = \tau$$

Joint Position Vector

Joint Velocity Vector

Joint Acceleration Vector

Joint Torque/Force Vector

Mass Matrix

Coriolis/Centrifugal Force Vector

Gravity Vector

Computed Torque Control

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + g(\theta) = \tau$$

The model-based portion of the control appears in a control law of the form:

$$\tau = \alpha v + \beta$$

where α and β are functions to be chosen so that if v is taken as the *new input* to the system, *the system appears to be a unit mass*.

$$\alpha = M(\theta)$$

$$\beta = C(\theta, \dot{\theta})\dot{\theta} + g(\theta)$$

Substituting the above control law into the dynamic equation, we get:

$$\ddot{\theta} = v$$

Servo Law Design

PD-Computed Torque

The servo law for the multi-input, multi-output robot system becomes

$$v = \ddot{\theta}_d + K_v \dot{E} + K_p E$$

where θ_d , $\dot{\theta}_d$ and $\ddot{\theta}_d$ are vectors of desired joint angles, velocity and accelerations respectively, K_v and K_p are now $n \times n$ matrices which are in general chosen to be diagonal. $E = \theta_d - \theta$, $\dot{E} = \dot{\theta}_d - \dot{\theta}$ are $n \times 1$ vectors of errors in position and velocity respectively.

Now, the error equation becomes:

$$\ddot{E} + K_v \dot{E} + K_p E = 0$$

$$\ddot{e}_i + k_{vi} \dot{e}_i + k_{pi} e_i = 0$$

PID-Computed Torque

With the addition of an integral term in the servo control law:

$$v = \ddot{\theta}_d + K_v \dot{E} + K_p E + K_I \int E dt$$

$$E = \theta_d - \theta$$

the closed-loop system becomes:

$$\dot{E} = \dot{\theta}_d - \dot{\theta}$$

$$\ddot{E} + K_v \dot{E} + K_p E + K_I \int E dt = 0$$

Gain Selection

Hence, the closed-loop characteristics equation is,

$$s^2 + k_{vi}s + k_{pi} = 0.$$

The characteristic equation can be expressed in the standard form as,

$$s^2 + 2\zeta \omega_n s + \omega_n^2 = \cancel{s^2 + 12s + 36} = 0$$

In practice, we must be careful not to excite structural oscillation and resonance of the joint.

$$\omega_n \leq 0.5 \omega_{res} = \cancel{0.5 \times 12} = 6 \text{ rad s}^{-1}$$

We choose

$$\zeta = 1$$

for a critically damped response.

Now, the error equation becomes:

$$\ddot{E} + K_v \dot{E} + K_p E = 0$$

$$\ddot{e}_i + k_{vi} \dot{e} + k_{pi} e = 0$$

$$k_{vi} = 2\zeta \omega_n$$

$$k_{pi} = \omega_n^2$$

Hybrid Position/Force Control

3.1 Hybrid Position/Force Control

In force control applications, we normally need to control the motions in certain directions (**motion control subspace**) and the force in the direction orthogonal to the motions (**force control subspace**).

$$v = \begin{bmatrix} v_T \\ v_N \end{bmatrix}$$

PD-Computed Torque

The servo law for the multi-input, multi-output robot system becomes

$$v = \ddot{\theta}_d + K_v \dot{E} + K_p E$$

Force Control

The normal force exerted on the surface is given by:

$$f = k_e (x - x_e) \quad \frac{1}{k_e} \ddot{f} = v_1$$

Let $e_f = f_d - f$, the force servo controller is:

$$v_1 = \frac{1}{k_e} (\ddot{f}_d + k_{1v} \dot{e}_f + k_{1p} e_f)$$

Thus

$$\ddot{e}_f + k_{1v} \dot{e}_f + k_{1p} e_f = 0$$

Hybrid Position/Force Control

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In force control applications, we normally need to control the motions in certain directions (**motion control subspace**) and the force in the direction orthogonal to the motions (**force control subspace**).

$$v = \begin{bmatrix} v_T \\ v_N \end{bmatrix}$$

Motion control

The motion servo law:

$$F_T = \ddot{x}_{Td} + K_{Tv}\dot{E}_T + K_{Tp}E_T$$

Let $e_f = f_d - f$, the force servo controller is:

$$v_1 = \frac{1}{k_e}(\ddot{f}_d + k_{1v}\dot{e}_f + k_{1p}e_f)$$

Thus

$$\ddot{e}_f + k_{1v}\dot{e}_f + k_{1p}e_f = 0$$

summary

Robot Control

< position control
force control

$\rightarrow x_d - x = e_p \rightarrow 0$
 $\Rightarrow$ design separately
 $\rightarrow f_d - f = e_f \rightarrow 0$

$$M \ddot{x} + C + g + f_e = \tau$$

position control: $F_T = \ddot{x}_{Td} + K_{Tv} \dot{E}_T + K_{Tp} E_T$

$$\ddot{E}_T + K_{Tv} \dot{E}_T + K_{Tp} E_T = 0$$

Force control $f_e = \underbrace{K_e}_{\text{stiffness matrix}} (x_N - x_e) \quad \ddot{x}_N = K_e^{-1} f_e$

$$F_N = K_e^{-1} (\ddot{f}_{ed} + K_{Nv} \dot{E}_N + K_{Np} E_N)$$

error dynamics: $\ddot{e}_f + \underbrace{K_{vp}}_{K_{vf}} \dot{e}_f + \underbrace{K_{pp}}_{K_{pf}} e_f = 0$

$$\begin{cases} K_v = 2\zeta \omega_n \\ K_p = \omega_n^2 \end{cases} \quad \begin{cases} \zeta \geq 1 \leftarrow \text{damping ratio} \\ \omega_n \leq 0.5 \omega_{res} \end{cases}$$

$$u = \alpha v + \beta$$

21-22 S2 Q2

2. The dynamic equations of a Cartesian robot which is in contact with a frictionless surface are given as follows:

$$2\ddot{x}_1 + 5x_1\dot{x}_1 + f_1 = u_1$$

$$10\ddot{x}_2 + 7x_2\dot{x}_2 + f_2 = u_2$$

$$5\ddot{x}_3 + 3x_3\dot{x}_3 + 49 = u_3$$

where x_1, x_2, x_3 represent the position of the end effector in Cartesian coordinates, f_1 and f_2 are the contact forces, u_1, u_2, u_3 are the control inputs. The contact forces exerted on the environment are given by

$$\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} 20 & 0 \\ 0 & 30 \end{bmatrix} \begin{bmatrix} x_1 - 1 \\ x_2 - 1 \end{bmatrix}$$

The system possesses unmodelled resonances at 8 rad/s, 16 rad/s and 24 rad/s.

- (a) Design a hybrid position and force controller for the robot. The system should be critically damped and does not excite all the unmodelled resonances.
(12 Marks)
- (b) The controller designed in part (a) is now implemented on the robot but a constant steady state position error is found on x_3 . Explain the possible effects and derive the error equations. Design a controller so that the steady state error can be eliminated and the same performance in part (a) is achieved.

(8 Marks)

The dynamic equations of a Cartesian robot which is in contact with a frictionless surface are given as follows:

$$2\ddot{x}_1 + 5x_1\dot{x}_1 + f_1 = u_1$$

$$10\ddot{x}_2 + 7x_2\dot{x}_2 + f_2 = u_2$$

$$5\ddot{x}_3 + 3x_3\dot{x}_3 + 49 = u_3$$

where x_1, x_2, x_3 represent the position of the end effector in Cartesian coordinates, f_1 and f_2 are the contact forces, u_1, u_2, u_3 are the control inputs. The contact forces exerted on the environment are given by

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(8 Marks)

Problem Setup:

A 3-DOF Cartesian robot (x_1, x_2, x_3) interacts with a frictionless surface.

• Dynamics Equations:

$$\begin{aligned} 2\ddot{x}_1 + 5x_1\dot{x}_1 + f_1 &= u_1 \\ 10\ddot{x}_2 + 7x_2\dot{x}_2 + f_2 &= u_2 \\ 5\ddot{x}_3 + 3x_3\dot{x}_3 + 49 &= u_3 \end{aligned}$$

• Contact Forces: Depend on x_1 and x_2 :

$$\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} 20 & 0 \\ 0 & 30 \end{bmatrix} \begin{bmatrix} x_1 - 1 \\ x_2 - 1 \end{bmatrix}$$

This implies:

- Stiffnesses: $k_{e1} = 20$ N/m, $k_{e2} = 30$ N/m
- Static surface positions: $x_{e1} = 1$ m, $x_{e2} = 1$ m

• Control Modes:

- x_1 : Force Control (since f_1 depends on x_1)
- x_2 : Force Control (since f_2 depends on x_2)
- x_3 : Position Control (no contact force f_3 mentioned)

• Parameters:

- Resonances: $\omega_{res} = \{8, 16, 24\}$ rad/s ($\omega_{res,min} = 8$ rad/s)
- Performance (Part a): Critically damped ($\zeta = 1$) for all axes.
- Gravity term in x_3 equation: 49.

Problem Setup:

A 3-DOF Cartesian robot (x_1, x_2, x_3) interacts with a frictionless surface.

- Dynamics Equations:**

$$\begin{aligned} 2\ddot{x}_1 + 5x_1\dot{x}_1 + f_1 &= u_1 \\ 10\ddot{x}_2 + 7x_2\dot{x}_2 + f_2 &= u_2 \\ 5\ddot{x}_3 + 3x_3\dot{x}_3 + 49 &= u_3 \end{aligned}$$

Write dynamics in matrix form: $M\ddot{X} + C(X, \dot{X}) + g(X) + F_c = U$, where $X = [x_1, x_2, x_3]^T$
 $U = [u_1, u_2, u_3]^T$, $F_c = [f_1, f_2, 0]^T$.

- Contact Forces:** Depend on x_1 and x_2 :

$$\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} 20 & 0 \\ 0 & 30 \end{bmatrix} \begin{bmatrix} x_1 - 1 \\ x_2 - 1 \end{bmatrix}$$

This implies:

- Stiffnesses: $k_{e1} = 20$ N/m, $k_{e2} = 30$ N/m
- Static surface positions: $x_{e1} = 1$ m, $x_{e2} = 1$ m

- Control Modes:**

- x_1 : Force Control (since f_1 depends on x_1)
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- x_3 : Position Control (no contact force f_3 mentioned)

- Parameters:**

- Resonances: $\omega_{res} = \{8, 16, 24\}$ rad/s ($\omega_{res, min} = 8$ rad/s)
- Performance (Part a): Critically damped ($\zeta = 1$) for all axes.
- Gravity term in x_3 equation: 49.

$$\underbrace{\begin{bmatrix} 2 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 5 \end{bmatrix}}_M \underbrace{\begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{bmatrix}}_{\ddot{X}} + \underbrace{\begin{bmatrix} 5x_1\dot{x}_1 \\ 7x_2\dot{x}_2 \\ 3x_3\dot{x}_3 \end{bmatrix}}_{C(X, \dot{X})} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 49 \end{bmatrix}}_{g(X)} + \underbrace{\begin{bmatrix} f_1 \\ f_2 \\ 0 \end{bmatrix}}_{F_c} = \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}}_U$$

Design the control law $U = \alpha v + \beta$ with:

- $\alpha = M = \text{diag}(2, 10, 5)$

- $\beta = C(X, \dot{X}) + g(X) + F_c = \begin{bmatrix} 5x_1\dot{x}_1 + f_1 \\ 7x_2\dot{x}_2 + f_2 \\ 3x_3\dot{x}_3 + 49 \end{bmatrix}$

Problem Setup:

A 3-DOF Cartesian robot (x_1, x_2, x_3) interacts with a frictionless surface.

• **Dynamics Equations:**

$$\begin{aligned} 2\ddot{x}_1 + 5x_1\dot{x}_1 + f_1 &= u_1 \\ 10\ddot{x}_2 + 7x_2\dot{x}_2 + f_2 &= u_2 \\ 5\ddot{x}_3 + 3x_3\dot{x}_3 + 49 &= u_3 \end{aligned}$$

• **Contact Forces:** Depend on x_1 and x_2 :

$$\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} 20 & 0 \\ 0 & 30 \end{bmatrix} \begin{bmatrix} x_1 - 1 \\ x_2 - 1 \end{bmatrix}$$

This implies:

- Stiffnesses: $k_{e1} = 20$ N/m, $k_{e2} = 30$ N/m
 - Static surface positions: $x_{e1} = 1$ m, $x_{e2} = 1$ m
- **Control Modes:**
- x_1 : Force Control (since f_1 depends on x_1)
 - x_2 : Force Control (since f_2 depends on x_2)
 - x_3 : Position Control (no contact force f_3 mentioned)

• **Parameters:**

- Resonances: $\omega_{res} = \{8, 16, 24\}$ rad/s ($\omega_{res,min} = 8$ rad/s)
- Performance (Part a): Critically damped ($\zeta = 1$) for all axes.
- Gravity term in x_3 equation: 49.

X1-Axis (Force Control):

- Error: $e_{f1} = f_{1d} - f_1$
- Desired Damping: $\zeta_{f1} = 1$
- Dynamics Relation: $\ddot{f}_1 = k_{e1}\ddot{x}_1 = 20v_1 \implies v_1 = \frac{1}{20}\ddot{f}_1$
- Servo Law (PD): $v_1 = \frac{1}{k_{e1}}(\ddot{f}_{1d} + k_{vf1}\dot{e}_{f1} + k_{pf1}e_{f1}) = \frac{1}{20}(\ddot{f}_{1d} + k_{vf1}\dot{e}_{f1} + k_{pf1}e_{f1})$
- Error Dynamics: $\ddot{e}_{f1} + k_{vf1}\dot{e}_{f1} + k_{pf1}e_{f1} = 0$
- Gain Selection:
 - Avoid resonance: $\omega_{nf1} \leq 0.5\omega_{res,min} = 0.5 \times 8 = 4$ rad/s. Choose $\omega_{nf1} = 4$.
 - $k_{pf1} = \omega_{nf1}^2 = 4^2 = 16$
 - $k_{vf1} = 2\zeta_{f1}\omega_{nf1} = 2 \times 1 \times 4 = 8$
- Resulting v_1 : $v_1 = \frac{1}{20}(\ddot{f}_{1d} + 8\dot{e}_{f1} + 16e_{f1})$

Problem Setup:

A 3-DOF Cartesian robot (x_1, x_2, x_3) interacts with a frictionless surface.

• **Dynamics Equations:**

$$\begin{aligned} 2\ddot{x}_1 + 5x_1\dot{x}_1 + f_1 &= u_1 \\ 10\ddot{x}_2 + 7x_2\dot{x}_2 + f_2 &= u_2 \\ 5\ddot{x}_3 + 3x_3\dot{x}_3 + 49 &= u_3 \end{aligned}$$

• **Contact Forces:** Depend on x_1 and x_2 :

$$\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} 20 & 0 \\ 0 & 30 \end{bmatrix} \begin{bmatrix} x_1 - 1 \\ x_2 - 1 \end{bmatrix}$$

This implies:

- Stiffnesses: $k_{e1} = 20$ N/m, $k_{e2} = 30$ N/m
 - Static surface positions: $x_{e1} = 1$ m, $x_{e2} = 1$ m
- **Control Modes:**
- x_1 : Force Control (since f_1 depends on x_1)
 - x_2 : Force Control (since f_2 depends on x_2)
 - x_3 : Position Control (no contact force f_3 mentioned)
- **Parameters:**
- Resonances: $\omega_{res} = \{8, 16, 24\}$ rad/s ($\omega_{res,min} = 8$ rad/s
 - Performance (Part a): Critically damped ($\zeta = 1$) for all axes.
 - Gravity term in x_3 equation: 49.

X2-Axis (Force Control):

- Error: $e_{f2} = f_{2d} - f_2$
- Desired Damping: $\zeta_{f2} = 1$
- Dynamics Relation: $\ddot{f}_2 = k_{e2}\ddot{x}_2 = 30v_2 \implies v_2 = \frac{1}{30}\ddot{f}_2$
- Servo Law (PD): $v_2 = \frac{1}{k_{e2}}(\ddot{f}_{2d} + k_{vf2}\dot{e}_{f2} + k_{pf2}e_{f2}) = \frac{1}{30}(\ddot{f}_{2d} + k_{vf2}\dot{e}_{f2} + k_{pf2}e_{f2})$
- Error Dynamics: $\ddot{e}_{f2} + k_{vf2}\dot{e}_{f2} + k_{pf2}e_{f2} = 0$
- Gain Selection:
 - Avoid resonance: $\omega_{nf2} \leq 0.5\omega_{res,min} = 4$ rad/s. Choose $\omega_{nf2} = 4$.
 - $k_{pf2} = \omega_{nf2}^2 = 4^2 = 16$
 - $k_{vf2} = 2\zeta_{f2}\omega_{nf2} = 2 \times 1 \times 4 = 8$
- Resulting v_2 : $v_2 = \frac{1}{30}(\ddot{f}_{2d} + 8\dot{e}_{f2} + 16e_{f2})$

Problem Setup:

A 3-DOF Cartesian robot (x_1, x_2, x_3) interacts with a frictionless surface.

• **Dynamics Equations:**

$$\begin{aligned} 2\ddot{x}_1 + 5x_1\dot{x}_1 + f_1 &= u_1 \\ 10\ddot{x}_2 + 7x_2\dot{x}_2 + f_2 &= u_2 \\ 5\ddot{x}_3 + 3x_3\dot{x}_3 + 49 &= u_3 \end{aligned}$$

• **Contact Forces:** Depend on x_1 and x_2 :

$$\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} 20 & 0 \\ 0 & 30 \end{bmatrix} \begin{bmatrix} x_1 - 1 \\ x_2 - 1 \end{bmatrix}$$

This implies:

- Stiffnesses: $k_{e1} = 20$ N/m, $k_{e2} = 30$ N/m
- Static surface positions: $x_{e1} = 1$ m, $x_{e2} = 1$ m
- **Control Modes:**
 - x_1 : Force Control (since f_1 depends on x_1)
 - x_2 : Force Control (since f_2 depends on x_2)
 - x_3 : Position Control (no contact force f_3 mentioned)
- **Parameters:**
 - Resonances: $\omega_{res} = \{8, 16, 24\}$ rad/s ($\omega_{res,min} = 8$ rad/s)
 - Performance (Part a): Critically damped ($\zeta = 1$) for all axes.
 - Gravity term in x_3 equation: 49.

X3-Axis (Position Control):

- Error: $e_{p3} = x_{3d} - x_3$
- Desired Damping: $\zeta_{p3} = 1$
- Servo Law (PD): $v_3 = \ddot{x}_{3d} + k_{vp3}\dot{e}_{p3} + k_{pp3}e_{p3}$
- Error Dynamics: $\ddot{e}_{p3} + k_{vp3}\dot{e}_{p3} + k_{pp3}e_{p3} = 0$
- Gain Selection:
 - Avoid resonance: $\omega_{np3} \leq 0.5\omega_{res,min} = 4$ rad/s. Choose $\omega_{np3} = 4$.
 - $k_{pp3} = \omega_{np3}^2 = 4^2 = 16$
 - $k_{vp3} = 2\zeta_{p3}\omega_{np3} = 2 \times 1 \times 4 = 8$
- Resulting v_3 : $v_3 = \ddot{x}_{3d} + 8\dot{e}_{p3} + 16e_{p3}$

Resulting $v_1 : v_1 = \frac{1}{20}(\ddot{f}_{1d} + 8\dot{e}_{f1} + 16e_{f1})$

Resulting $v_3 : v_3 = \ddot{x}_{3d} + 8\dot{e}_{p3} + 16e_{p3}$

Resulting $v_2 : v_2 = \frac{1}{30}(\ddot{f}_{2d} + 8\dot{e}_{f2} + 16e_{f2})$

$$\begin{aligned} 2\ddot{x}_1 + 5x_1\dot{x}_1 + f_1 &= u_1 \\ 10\ddot{x}_2 + 7x_2\dot{x}_2 + f_2 &= u_2 \\ 5\ddot{x}_3 + 3x_3\dot{x}_3 + 49 &= u_3 \end{aligned}$$

Step 3: Final Controller Assembly (Part a)

Substitute v_1, v_2, v_3 into $U = \alpha v + \beta$:

$$u_1 = 2v_1 + 5x_1\dot{x}_1 + f_1 = 2 \left[\frac{1}{20}(\ddot{f}_{1d} + 8\dot{e}_{f1} + 16e_{f1}) \right] + 5x_1\dot{x}_1 + f_1$$

$$u_1 = \frac{1}{10}(\ddot{f}_{1d} + 8\dot{e}_{f1} + 16e_{f1}) + 5x_1\dot{x}_1 + f_1$$

$$u_2 = 10v_2 + 7x_2\dot{x}_2 + f_2 = 10 \left[\frac{1}{30}(\ddot{f}_{2d} + 8\dot{e}_{f2} + 16e_{f2}) \right] + 7x_2\dot{x}_2 + f_2$$

$$u_2 = \frac{1}{3}(\ddot{f}_{2d} + 8\dot{e}_{f2} + 16e_{f2}) + 7x_2\dot{x}_2 + f_2$$

$$u_3 = 5v_3 + 3x_3\dot{x}_3 + 49 = 5(\ddot{x}_{3d} + 8\dot{e}_{p3} + 16e_{p3}) + 3x_3\dot{x}_3 + 49$$

where $e_{f1} = f_{1d} - f_1$, $e_{f2} = f_{2d} - f_2$, and $e_{p3} = x_{3d} - x_3$. Also, $f_1 = 20(x_1 - 1)$ and $f_2 = 30(x_2 - 1)$.

summary

Robot Control

position control
force control

$x_d - x = e_p \rightarrow 0$
 $\Rightarrow$ design separately
 $f_d - f = e_f \rightarrow 0$

$$M \ddot{x} + C + g + f_e = \tau$$

position control: $F_T = \ddot{x}_{Td} + K_{Tv} \dot{E}_T + K_{Tp} \cdot E_T$

$$\ddot{E}_T + K_{Tv} \dot{E}_T + K_{Tp} E_T = 0$$

Force control $f_e = K_e (x_N - x_e)$ $\ddot{x}_N = K_e^{-1} f_e$
 $\downarrow$
 stiffness matrix

$$F_N = K_e^{-1} (\ddot{f}_{ed} + K_{Nv} \dot{E}_N + K_{Np} \cdot E_N)$$

error dynamics: $\ddot{e}_f + \underbrace{K_{vp}}_{K_{vf}} \dot{e}_f + \underbrace{K_{pp}}_{K_{pf}} e_f = 0$

$$\begin{cases} K_v = 2 \zeta \omega_n \\ K_p = \omega_n^2 \end{cases} \begin{cases} \zeta \geq 1 \leftarrow \text{damping ratio} \\ \omega_n \leq 0.5 \omega_{res} \end{cases}$$

$$u = \alpha v + \beta$$

23-24 S2 Q2

2. A Cartesian robot with two degrees of freedom is in contact with a workpiece, which is shown in Figure 2.

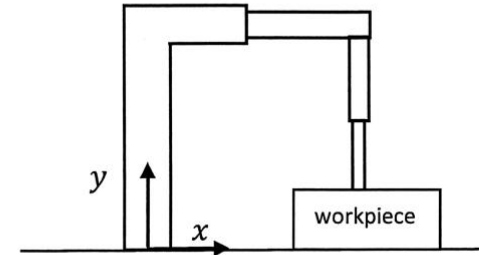


Figure 2

The dynamic equations of the robot, when it is not in contact with the workpiece, are given as follows:

$$5\ddot{x} + 10x\dot{x} + d_x = u_x$$

$$10\ddot{y} + 15y\dot{y} + 98 + d_y = u_y$$

where u_x, u_y are the control inputs and d_x, d_y are the unknown constant disturbances. The stiffness of the workpiece in any direction is given as 50 N/m and the static position of the workpiece in the y axis is 0.25 m. The system possesses unmodelled resonances at 7 rad/s and 15 rad/s.

- (a) Assume that $d_x = 0$ and $d_y = 0$. Design a hybrid position and force controller for the robot. The motion control subspace should be critically damped, and the force control subspace should be overdamped with a damping ratio of 1.25. The control gains should be chosen to be as high as possible and the system should not excite all the unmodelled resonances.

(13 Marks)

- (b) Assume that d_x, d_y are not negligible. Design an appropriate hybrid position and force controller and show that the steady state error can be eliminated. In case that the stiffness of the workpiece is wrongly estimated as 49 N/m, discuss its effects on the controller.

(7 Marks)

2. A Cartesian robot with two degrees of freedom is in contact with a workpiece, which is shown in Figure 2.

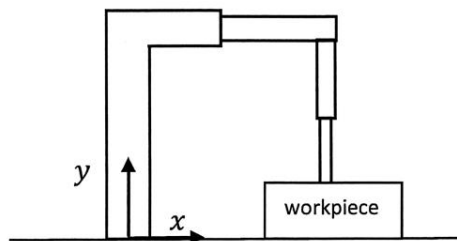


Figure 2

The dynamic equations of the robot, when it is not in contact with the workpiece, are given as follows:

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(13 Marks)

- (b) Assume that d_x, d_y are not negligible. Design an appropriate hybrid position and force controller and show that the steady state error can be eliminated. In case that the stiffness of the workpiece is wrongly estimated as 49 N/m, discuss its effects on the controller.

(7 Marks)

Problem Setup:

A 2-DOF Cartesian robot interacts with a workpiece (Figure 2).

- **Contact:** The robot contacts the workpiece along the vertical y -axis.
- **Control Modes:**
 - x -axis: **Position** Control (Motion Control Subspace)
 - y -axis: **Force** Control (Force Control Subspace)
- **Dynamic Equations:**

$$5\ddot{x} + 10x\dot{x} + d_x = u_x$$

$$10\ddot{y} + 15y\dot{y} + 98 + d_y = u_y$$
- **Parameters:**
 - Workpiece Stiffness: $k_e = 50$ N/m (acting in y -direction)
 - Workpiece Static Position: $y_e = 0.25$ m
 - Unmodelled Resonances: $\omega_{res1} = 7$ rad/s, $\omega_{res2} = 15$ rad/s ($\omega_{res,min} = 7$ rad/s)

2. A Cartesian robot with two degrees of freedom is in contact with a workpiece, which is shown in Figure 2.

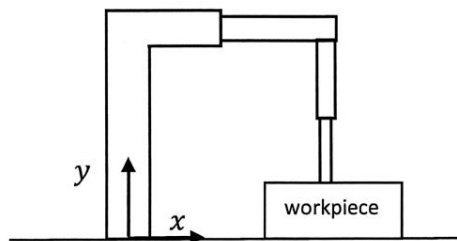


Figure 2

The dynamic equations of the robot, when it is not in contact with the workpiece, are given as follows:

$$5\ddot{x} + 10x\dot{x} + d_x = u_x$$

$$10\ddot{y} + 15y\dot{y} + 98 + d_y = u_y$$

where u_x, u_y are the control inputs and d_x, d_y are the unknown constant disturbances. The stiffness of the workpiece in any direction is given as 50 N/m and the static position of the workpiece in the y axis is 0.25 m. The system possesses unmodelled resonances at 7 rad/s and 15 rad/s.

- (a) Assume that $d_x = 0$ and $d_y = 0$. Design a hybrid position and force controller for the robot. The motion control subspace should be critically damped, and the force control subspace should be overdamped with a damping ratio of 1.25. The control gains should be chosen to be as high as possible and the system should not excite all the unmodelled resonances.

(13 Marks)

- (b) Assume that d_x, d_y are not negligible. Design an appropriate hybrid position and force controller and show that the steady state error can be eliminated. In case that the stiffness of the workpiece is wrongly estimated as 49 N/m, discuss its effects on the controller.

(7 Marks)

Step 1: Computed Torque Control (Linearization)

Write dynamics in matrix form: $M\ddot{X} + C(X, \dot{X}) + g(X) = \tau$, where $X = [x, y]^T$, $\tau = [u_x, u_y]^T$.

$$\underbrace{\begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix}}_M \underbrace{\begin{bmatrix} \ddot{x} \\ \ddot{y} \end{bmatrix}}_{\ddot{X}} + \underbrace{\begin{bmatrix} 10x\dot{x} \\ 15y\dot{y} \end{bmatrix}}_{C(X, \dot{X})} + \underbrace{\begin{bmatrix} 0 \\ 98 \end{bmatrix}}_{g(X)} = \underbrace{\begin{bmatrix} u_x \\ u_y \end{bmatrix}}_{\tau}$$

Design the control law $\tau = \alpha v + \beta$ with:

- $\alpha = M = \begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix}$
- $\beta = C(X, \dot{X}) + g(X) = \begin{bmatrix} 10x\dot{x} \\ 15y\dot{y} + 98 \end{bmatrix}$

Substituting this into the dynamics yields the linearized system:

$$v = \begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} \ddot{x} \\ \ddot{y} \end{bmatrix}$$

2. A Cartesian robot with two degrees of freedom is in contact with a workpiece, which is shown in Figure 2.

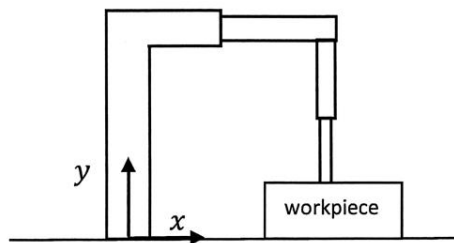


Figure 2

The dynamic equations of the robot, when it is not in contact with the workpiece, are given as follows:

$$5\ddot{x} + 10x\dot{x} + d_x = u_x$$

$$10\ddot{y} + 15y\dot{y} + 98 + d_y = u_y$$

where u_x, u_y are the control inputs and d_x, d_y are the unknown constant disturbances. The stiffness of the workpiece in any direction is given as 50 N/m and the static position of the workpiece in the y axis is 0.25 m. The system possesses unmodelled resonances at 7 rad/s and 15 rad/s.

- (a) Assume that $d_x = 0$ and $d_y = 0$. Design a hybrid position and force controller for the robot. The motion control subspace should be critically damped, and the force control subspace should be overdamped with a damping ratio of 1.25. The control gains should be chosen to be as high as possible and the system should not excite all the unmodelled resonances.

(13 Marks)

- (b) Assume that d_x, d_y are not negligible. Design an appropriate hybrid position and force controller and show that the steady state error can be eliminated. In case that the stiffness of the workpiece is wrongly estimated as 49 N/m, discuss its effects on the controller.

(7 Marks)

Step 2: Servo Controller Design (PD)

• X-Axis (Position Control):

- **Error:** $e_x = x_d - x$
- **Servo Law (PD):** $v_x = \ddot{x}_d + k_{vx}\dot{e}_x + k_{px}e_x$
- **Linearization:** $v_x = \ddot{x}$

We can get **Error Equation** from the 3 equations above: $\ddot{e}_x + k_{vx}\dot{e}_x + k_{px}e_x = 0$

- **Characteristic Equation:** $s^2 + k_{vx}s + k_{px} = 0$
- **Desired Damping:** $\zeta_x = 1$
- **Gain Selection:**
 - Avoid **resonance:** $\omega_{nx} \leq 0.5\omega_{res,min} = 0.5 \times 7 = 3.5$ rad/s. Choose $\omega_{nx} = 3.5$ rad/s.
 - $k_{px} = \omega_{nx}^2 = (3.5)^2 = 12.25$
 - $k_{vx} = 2\zeta_x\omega_{nx} = 2 \times 1 \times 3.5 = 7$
- **Resulting** v_x : $v_x = \ddot{x}_d + 7\dot{e}_x + 12.25e_x$

2. A Cartesian robot with two degrees of freedom is in contact with a workpiece, which is shown in Figure 2.

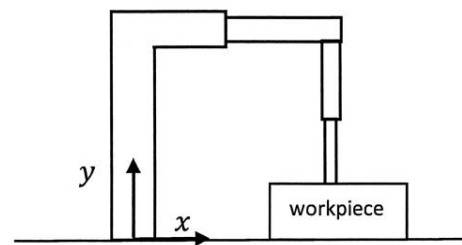


Figure 2

The dynamic equations of the robot, when it is not in contact with the workpiece, are given as follows:

$$5\ddot{x} + 10x\dot{x} + d_x = u_x$$

$$10\ddot{y} + 15y\dot{y} + 98 + d_y = u_y$$

where u_x, u_y are the control inputs and d_x, d_y are the unknown constant disturbances. The stiffness of the workpiece in any direction is given as 50 N/m and the static position of the workpiece in the y axis is 0.25 m. The system possesses unmodelled resonances at 7 rad/s and 15 rad/s.

- Assume that $d_x = 0$ and $d_y = 0$. Design a hybrid position and force controller for the robot. The motion control subspace should be critically damped, and the force control subspace should be overdamped with a damping ratio of 1.25. The control gains should be chosen to be as high as possible and the system should not excite all the unmodelled resonances.
- Assume that d_x, d_y are not negligible. Design an appropriate hybrid position and force controller and show that the steady state error can be eliminated. In case that the stiffness of the workpiece is wrongly estimated as 49 N/m, discuss its effects on the controller.

Y-Axis (Force Control):

- Force-Position Relation:** $f = k_e(y - y_e) = 50(y - 0.25)$
- Dynamics Relation:** $\ddot{f} = k_e\ddot{y} = k_e v_y \implies v_y = \frac{1}{k_e}\ddot{f} = \frac{1}{50}\ddot{f}$
- Error:** $e_f = f_d - f$
- Servo Law (PD):** $v_y = \frac{1}{k_e}(\ddot{f}_d + k_{vf}\dot{e}_f + k_{pf}e_f) = \frac{1}{50}(\ddot{f}_d + k_{vf}\dot{e}_f + k_{pf}e_f)$
- Linearization:** $v_y = \ddot{y} = \frac{1}{50}\ddot{f}$
- Error Equation:** $\ddot{e}_f + k_{vf}\dot{e}_f + k_{pf}e_f = 0$
- Characteristic Equation:** $s^2 + k_{vf}s + k_{pf} = 0$
- Desired Damping:** $\zeta_f = 1.25$
- Gain Selection:**
 - Avoid resonance: $\omega_{nf} \leq 0.5\omega_{res,min} = 3.5$ rad/s. Choose $\omega_{nf} = 3.5$ rad/s.
 - $k_{pf} = \omega_{nf}^2 = (3.5)^2 = 12.25$
 - $k_{vf} = 2\zeta_f\omega_{nf} = 2 \times 1.25 \times 3.5 = 8.75$
- Resulting v_y :** $v_y = \frac{1}{50}(\ddot{f}_d + 8.75\dot{e}_f + 12.25e_f)$

(7 Marks)

Step 1: Computed Torque Control (Linearization)

Write dynamics in matrix form: $M\ddot{X} + C(X, \dot{X}) + g(X) = \tau$, where $X = [x, y]^T$, $\tau = [u_x, u_y]^T$.

$$\underbrace{\begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix}}_M \underbrace{\begin{bmatrix} \ddot{x} \\ \ddot{y} \end{bmatrix}}_{\ddot{X}} + \underbrace{\begin{bmatrix} 10x\dot{x} \\ 15y\dot{y} \end{bmatrix}}_{C(X, \dot{X})} + \underbrace{\begin{bmatrix} 0 \\ 98 \end{bmatrix}}_{g(X)} = \underbrace{\begin{bmatrix} u_x \\ u_y \end{bmatrix}}_{\tau}$$

Design the control law $\tau = \alpha v + \beta$ with:

- $\alpha = M = \begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix}$
- $\beta = C(X, \dot{X}) + g(X) = \begin{bmatrix} 10x\dot{x} \\ 15y\dot{y} + 98 \end{bmatrix}$

Substituting this into the dynamics yields the linearized system:

$$v = \begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} \ddot{x} \\ \ddot{y} \end{bmatrix}$$

$$\text{Resulting } v_x : v_x = \ddot{x}_d + 7\dot{e}_x + 12.25e_x$$

$$\text{Resulting } v_y : v_y = \frac{1}{50}(\ddot{f}_d + 8.75\dot{e}_f + 12.25e_f)$$

Step 3: Final Controller Assembly (Part a)

Substitute v_x, v_y into $\tau = \alpha v + \beta$:

$$u_x = 5v_x + 10x\dot{x} = 5(\ddot{x}_d + 7\dot{e}_x + 12.25e_x) + 10x\dot{x}$$

$$u_y = 10v_y + 15y\dot{y} + 98 = 10 \left[\frac{1}{50}(\ddot{f}_d + 8.75\dot{e}_f + 12.25e_f) \right] + 15y\dot{y} + 98$$

$$u_y = \frac{1}{5}(\ddot{f}_d + 8.75\dot{e}_f + 12.25e_f) + 15y\dot{y} + 98$$

where $e_x = x_d - x$ and $e_f = f_d - f = f_d - 50(y - 0.25)$.

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + g(\theta) = \tau$$

24-25 s1 q2

2. The dynamic equations of a robot when it is in contact with a workpiece are given as follows:

$$(5q_2^2 + 1)\ddot{q}_1 + 10q_2\dot{q}_1\dot{q}_2 + 3\dot{q}_1 + 47.5 q_2 \cos(q_1) + d_1 = u_1$$

$$5\ddot{q}_2 - 5q_2\dot{q}_1^2 + 5\dot{q}_2 + 47.5 \sin(q_1) + d_2 = u_2$$

$$3\ddot{q}_3 + 2\dot{q}_3 + 29.4 + d_3 + f = u_3$$

where u_1, u_2, u_3 are the control inputs, q_1, q_2, q_3 are the joint variables, d_1, d_2, d_3 are the unknown disturbances and $f = 22(q_3 - 0.2)$ is the contact force. The system possesses unmodelled resonances at 5 rad/s, 10 rad/s and 15 rad/s.

- (a) If d_1, d_2, d_3 are zero, design a hybrid position and force controller for the robot to track the desired trajectories q_{1d}, q_{2d} and the desired force f_d . The motion control subspace should be overdamped with a damping ratio of 1.05, and the force control subspace should be critically damped. The control system should not excite all the unmodelled resonances.

(14 Marks)

- (b) If d_1, d_2, d_3 are not zero, derive the error equations of the system based on the hybrid position and force controller designed in part (a).

(6 Marks)

2. The dynamic equations of a robot when it is in contact with a workpiece are given as follows:

$$(5q_2^2 + 1) \ddot{q}_1 + 10q_2\dot{q}_1\dot{q}_2 + 3\dot{q}_1 + 47.5 q_2 \cos(q_1) + d_1 = u_1$$

$$5\ddot{q}_2 - 5q_2\dot{q}_1^2 + 5\dot{q}_2 + 47.5 \sin(q_1) + d_2 = u_2$$

$$3\ddot{q}_3 + 2\dot{q}_3 + 29.4 + d_3 + f = u_3$$

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + g(\theta) = \tau$$

where u_1, u_2, u_3 are the control inputs, q_1, q_2, q_3 are the joint variables, d_1, d_2, d_3 are the unknown disturbances and $f = 22(q_3 - 0.2)$ is the contact force. The system possesses unmodelled resonances at 5 rad/s, 10 rad/s and 15 rad/s.

- (a) If d_1, d_2, d_3 are zero, design a hybrid position and force controller for the robot to track the desired trajectories q_{1d}, q_{2d} and the desired force f_d . The motion control subspace should be overdamped with a damping ratio of 1.05, and the force control subspace should be critically damped. The control system should not excite all the unmodelled resonances.

(14 Marks)

The whole controller:

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 5q_2^2 + 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} + \begin{bmatrix} 10q_2\dot{q}_1\dot{q}_2 + 3\dot{q}_1 \\ -5q_2\dot{q}_1^2 + 5\dot{q}_2 \\ 2\dot{q}_3 \end{bmatrix} + \begin{bmatrix} 47.5q_2 \cos(q_1) \\ 47.5 \sin(q_1) \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 22(q_3 - 0.2) \end{bmatrix} + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$M(\theta)\ddot{\theta} \quad C(\theta, \dot{\theta})\dot{\theta} \quad g(\theta) \quad f(x) \quad d$

The whole controller:

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + g(\theta) = \tau$$

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 5q_2^2 + 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}}_{M(\theta)\ddot{\theta}} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} + \underbrace{\begin{bmatrix} 10q_2\dot{q}_1\dot{q}_2 + 3\dot{q}_1 \\ -51_2\dot{q}_1^2 + 5\dot{q}_2 \\ 2\dot{q}_3 \end{bmatrix}}_{C(\theta, \dot{\theta})\dot{\theta}} + \underbrace{\begin{bmatrix} 47.5q_2 \cos(q_1) \\ 47.5 \sin(q_1) \\ 0 \end{bmatrix}}_{g(\theta)} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 22(q_3 - 0.2) \end{bmatrix}}_{f(x)} + \underbrace{\begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}}_d$$

- (a) If d_1, d_2, d_3 are zero, design a hybrid position and force controller for the robot to track the desired trajectories q_{1d}, q_{2d} and the desired force f_d . The motion control subspace should be overdamped with a damping ratio of 1.05, and the force control subspace should be critically damped. The control system should not excite all the unmodelled resonances.

(14 Marks)

For motion control,

$$\xi_m = 1.05, \quad w_n = 0.5 \times \min w_{res} = 0.5 \times 5 = 2.5 \text{ rad/s}$$

$$k_v = 2\xi_m w_n = 5.25, \quad k_p = w_n^2 = 6.25$$

$$v_1 = \ddot{q}_1 + 5.25\dot{e}_1 + 6.25e_1$$

$$v_2 = \ddot{q}_2 + 5.25\dot{e}_2 + 6.25e_2$$

The whole controller:

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + g(\theta) = \tau$$

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 5q_2^2 + 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}}_{M(\theta)\ddot{\theta}} + \underbrace{\begin{bmatrix} 10q_2\dot{q}_1\dot{q}_2 + 3\dot{q}_1 \\ -51_2\dot{q}_1^2 + 5\dot{q}_2 \\ 2\dot{q}_3 \end{bmatrix}}_{C(\theta, \dot{\theta})\dot{\theta}} + \underbrace{\begin{bmatrix} 47.5q_2 \cos(q_1) \\ 47.5 \sin(q_1) \\ 0 \end{bmatrix}}_{g(\theta)} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 22(q_3 - 0.2) \end{bmatrix}}_{f(x)} + \underbrace{\begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}}_d$$

- (a) If d_1, d_2, d_3 are zero, design a hybrid position and force controller for the robot to track the desired trajectories q_{1d}, q_{2d} and the desired force f_d . The motion control subspace should be overdamped with a damping ratio of 1.05, and the force control subspace should be critically damped. The control system should not excite all the unmodelled resonances.

(14 Marks)

For force control,

$$\xi_f = 1, \quad w_n = 0.5 \times \min w_{res} = 0.5 \times 5 = 2.5 \text{ rad/s}$$

$$k_v = 2\xi_m w_n = 5, \quad k_p = w_n^2 = 6.25$$

$$v_3 = \frac{1}{22}(\ddot{q}_3 + 5\dot{e}_3 + 6.25e_3)$$

The whole controller:

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + g(\theta) = \tau$$

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 5q_2^2 + 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}}_{M(\theta)\ddot{\theta}} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} + \underbrace{\begin{bmatrix} 10q_2\dot{q}_1\dot{q}_2 + 3\dot{q}_1 \\ -51_2\dot{q}_1^2 + 5\dot{q}_2 \\ 2\dot{q}_3 \end{bmatrix}}_{C(\theta, \dot{\theta})\dot{\theta}} + \underbrace{\begin{bmatrix} 47.5q_2 \cos(q_1) \\ 47.5 \sin(q_1) \\ 0 \end{bmatrix}}_{g(\theta)} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 22(q_3 - 0.2) \end{bmatrix}}_{f(x)} + \underbrace{\begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}}_d$$

To be more detailed, take u_1 as an example:

$$u_1 = \alpha_1 v_1 + \beta_1$$

where

$$\alpha_1 = 5q_2^2 + 1$$

$$\beta_1 = 10q_2\dot{q}_1\dot{q}_2 + 3\dot{q}_1 + 47.5q_2 \cos(q_1) + d_1$$

The whole controller:

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + g(\theta) = \tau$$

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 5q_2^2 + 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}}_{M(\theta)\ddot{\theta}} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} + \underbrace{\begin{bmatrix} 10q_2\dot{q}_1\dot{q}_2 + 3\dot{q}_1 \\ -51_2\dot{q}_1^2 + 5\dot{q}_2 \\ 2\dot{q}_3 \end{bmatrix}}_{C(\theta, \dot{\theta})\dot{\theta}} + \underbrace{\begin{bmatrix} 47.5q_2 \cos(q_1) \\ 47.5 \sin(q_1) \\ 0 \end{bmatrix}}_{g(\theta)} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 22(q_3 - 0.2) \end{bmatrix}}_{f(x)} + \underbrace{\begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}}_d$$

Let $e_1 = q_{1d} - q_1$, we have $\ddot{e}_1 = \ddot{q}_{1d} - \ddot{q}_1$.

Control Law:

$$v_1 = \ddot{q}_{1d} + k_{v1}\dot{e}_1 + k_{p1}e_1$$

Substitute $\ddot{e}_1 = \ddot{q}_{1d} - \ddot{q}_1$ into

$$\ddot{q}_1 = v_1 = \ddot{q}_{1d} + k_{v1}\dot{e}_1 + k_{p1}e_1$$

We get error equation

$$\ddot{e}_1 + k_{v1}\dot{e}_1 + k_{p1}e_1 = 0$$

Use Laplace Transformation to error equation, we can get the closed-loop characteristics equation

$$s^2 + k_{v1}s + k_{p1} = 0$$

The whole controller:

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + g(\theta) = \tau$$

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 5q_2^2 + 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}}_{M(\theta)\ddot{\theta}} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} + \underbrace{\begin{bmatrix} 10q_2\dot{q}_1\dot{q}_2 + 3\dot{q}_1 \\ -51_2\dot{q}_1^2 + 5\dot{q}_2 \\ 2\dot{q}_3 \end{bmatrix}}_{C(\theta, \dot{\theta})\dot{\theta}} + \underbrace{\begin{bmatrix} 47.5q_2 \cos(q_1) \\ 47.5 \sin(q_1) \\ 0 \end{bmatrix}}_{g(\theta)} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 22(q_3 - 0.2) \end{bmatrix}}_{f(x)} + \underbrace{\begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}}_d$$

Use Laplace Transformation to error equation, we can get the closed-loop characteristics equation

$$s^2 + k_{v1}s + k_{p1} = 0$$

The characteristic equation of a second order system can be expressed in the standard form:

$$s^2 + 2\xi w_n \dot{e} + w_n^2 e = 0$$

Hence,

$$k_{v1} = 2\xi w_n, \quad k_{p1} = w_n^2$$

The whole controller:

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + g(\theta) = \tau$$

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 5q_2^2 + 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}}_{M(\theta)\ddot{\theta}} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} + \underbrace{\begin{bmatrix} 10q_2\dot{q}_1\dot{q}_2 + 3\dot{q}_1 \\ -51_2\dot{q}_1^2 + 5\dot{q}_2 \\ 2\dot{q}_3 \end{bmatrix}}_{C(\theta, \dot{\theta})\dot{\theta}} + \underbrace{\begin{bmatrix} 47.5q_2 \cos(q_1) \\ 47.5 \sin(q_1) \\ 0 \end{bmatrix}}_{g(\theta)} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 22(q_3 - 0.2) \end{bmatrix}}_{f(x)} + \underbrace{\begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}}_d$$

So, for u_1 , the controller is

$$u_1 = (5q_2^2 + 1)v_1 + (10q_2\dot{q}_1\dot{q}_2 + 3\dot{q}_1 + 47.5q_2 \cos(q_1) + d_1)$$

the control law is

$$v_1 = \ddot{q}_{1d} + k_{v1}\dot{e}_1 + k_{p1}e_1$$

where $k_{v1} = 2\xi w_n$, $k_{p1} = w_n^2$.

The dynamic equations of a robot when it is in contact with a workpiece are given as follows:

$$(5q_2^2 + 1)\ddot{q}_1 + 10q_2\dot{q}_1\dot{q}_2 + 3\dot{q}_1 + 47.5q_2\cos(q_1) + d_1 = u_1$$

$$5\ddot{q}_2 - 5q_2\dot{q}_1^2 + 5\dot{q}_2 + 47.5\sin(q_1) + d_2 = u_2$$

$$3\ddot{q}_3 + 2\dot{q}_3 + 29.4 + d_3 + f = u_3$$

where u_1, u_2, u_3 are the control inputs, q_1, q_2, q_3 are the joint variables, d_1, d_2, d_3 are the unknown disturbances and $f = 22(q_3 - 0.2)$ is the contact force. The system possesses unmodelled resonances at 5 rad/s, 10 rad/s and 15 rad/s.

- (a) If d_1, d_2, d_3 are zero, design a hybrid position and force controller for the robot to track the desired trajectories q_{1d}, q_{2d} and the desired force f_d . The motion control subspace should be overdamped with a damping ratio of 1.05, and the force control subspace should be critically damped. The control system should not excite all the unmodelled resonances.

(14 Marks)

- (b) If d_1, d_2, d_3 are not zero, derive the error equations of the system based on the hybrid position and force controller designed in part (a).

(6 Marks)

Q2.

$$\begin{bmatrix} 5q_2^2 + 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \\ \ddot{q}_3 \end{bmatrix} + \begin{bmatrix} 10q_2\dot{q}_1\dot{q}_2 + 3\dot{q}_1 + 47.5q_2\cos q_1 \\ -5q_2\dot{q}_1^2 + 5\dot{q}_2 + 47.5\sin q_1 \\ 29.4 + f \end{bmatrix} + \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$\alpha \quad \quad \quad \checkmark \quad + \quad \quad \quad \beta$

$$V = \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

for joint 1: motion control, $\xi = 1.05$ $\omega_{res} = 5$

$$\omega_{n1} \in 0.5 \omega_{res} = 2.5 \text{ rad/s}$$

$$K_{v1} = 2\xi\omega_{n1} = 5.25 \quad K_{p1} = \omega_{n1}^2 = 6.25$$

$$V_1 = \dot{q}_1 + 5.25\dot{e}_1 + 6.25e_1 = 0$$

joint 2: $\xi = 1.05$ $\omega_{res} = 10 \text{ rad/s}$

$$\omega_{n2} \in 5 \text{ rad/s} \quad K_{v2} = 2\xi\omega_{n2} = 10.5 \quad K_{p2} = \omega_{n2}^2 = 25$$

$$V_2 = \dot{q}_2 + 10.5\dot{e}_2 + 25e_2$$

joint 3: $\xi = 1$ $\omega_{res} = 5 \text{ rad/s}$ $K = 22$

$$\omega_{n3} \leq 2.5 \text{ rad/s} \quad K_{v3} = 5 \quad K_{p3} = 6.25$$

$$V_3 = \frac{1}{K} (\dot{q}_3 + K_{v3}\dot{e}_3 + K_{p3}e_3) = \frac{1}{22} (\dot{q}_3 + 5\dot{e}_3 + 6.25e_3)$$

$$\alpha V + \beta$$

Thank you